計算機組織 Assignment 1 參考解答

CO3020 Computer Organization　Chapter 1 and 2

1.

$$n=2019-2010=9, r=\left(\frac{V_{2019}}{V_{2010}}\right)^{1/9}-1, T_d=\frac{\ln 2}{\ln(1+r)}.$$

$$r_{Tech}=\left(\frac{32}{14}\right)^{2/9}-1\approx 20.17\%, T_{d,Tech}=\frac{9\ln 2}{2\ln(32/14)}\approx 3.77\ years.$$

Desktop Processor	Year	Tech	Max. Clock Speed (GHz)	Integer IPC/core	Cores	Max. DRAM Bandwidth (GB/s)	SP Floating Point (Gflop/s)	MiB
Westmere i7-620	2010	32	3.33	4	2	17.1	107	4
Ivy Bridge i7-3770K	2013	22	3.90	6	4	25.6	250	8
Broadwell i7-6700K	2015	14	4.20	8	4	34.1	269	8
Kaby Lake i7-7700K	2017	14	4.50	8	4	38.4	288	8
Coffee Lake i7-9700K	2019	14	4.90	8	8	42.7	627	12
Imp./year (%)		**20.17**	**4.39**	**8.01**	**16.65**	**10.70**	**21.71**	**12.98**
Doubles every (Y)		**3.77**	**16.15**	**9.00**	**4.50**	**6.82**	**3.53**	**5.68**

2.

(a)

$$T_{P1}=\frac{10^5(1)+2\times 10^5(2)+5\times 10^5(3)+2\times 10^5(3)}{2.5\times 10^9}=1.04\times 10^{-3}s.$$

$$T_{P2}=\frac{10^6(2)}{3\times 10^9}=6.6667\times 10^{-4}s.$$

$T_{P2}<T_{P1}$，P2 較快。

(b)

$$\overline{CPI}_{P1}=0.1(1)+0.2(2)+0.5(3)+0.2(3)=2.6.$$

$$\overline{CPI}_{P2}=0.1(2)+0.2(2)+0.5(2)+0.2(2)=2.0.$$

(c)

$$C_{P1}=10^6(2.6)=2.6\times 10^6 cycles.$$

$$C_{P2}=10^6(2.0)=2.0\times 10^6 cycles.$$

3.

(a)

$$f=\frac{1}{1\times10^{-9}}=10^{9}Hz, CPI=\frac{Tf}{IC}.$$

$$CPI_A=\frac{1.1(10^9)}{1.0\times10^9}=1.1, CPI_B=\frac{1.5(10^9)}{1.2\times10^9}=1.25.$$

(b)

$$\frac{IC_A CPI_A}{f_A}=\frac{IC_B CPI_B}{f_B}.$$

$$\frac{f_A}{f_B}=\frac{IC_A CPI_A}{IC_B CPI_B}=\frac{1.0\times10^9(1.1)}{1.2\times10^9(1.25)}=\frac{11}{15}\approx0.7333.$$

(c)

$$T_{new}=\frac{6.0\times10^8(1.1)}{10^9}=0.66\,s.$$

$$S_A=\frac{1.1}{0.66}\approx1.6667, S_B=\frac{1.5}{0.66}\approx2.2727.$$

4.

$$T_{INT}=250-70-85-40=55\,s.$$

(a)

$$T'_{FP}=70(0.8)=56\,s, T'=56+85+40+55=236\,s.$$

$$\Delta T=250-236=14\,s, \frac{14}{250}\times100\%=5.6\%.$$

(b)

$$T'=250(0.8)=200\,s, T'_{INT}=200-(70+85+40)=5\,s.$$

$$\Delta T_{INT}=55-5=50\,s, \frac{50}{55}\times100\%\approx90.91\%.$$

(c)

$$\Delta T_{max}=40\,s, \frac{40}{250}\times100\%=16\%<20\%.$$

$$T_{min}=250-40=210\,s>200\,s\Rightarrow \text{不可達成}.$$

5.

$$C=[50(1)+110(1)+80(4)+16(2)]\times 10^6=512\times 10^6 cycles.$$

$$T=\frac{512\times 10^6}{2\times 10^9}=0.256s, C_{target}=256\times 10^6.$$

(a)

$$50x+110+320+32=256.$$

$$CPI'_{FP}=x=\frac{256-462}{50}=-4.12<0\Rightarrow 不可達成.$$

(b)

$$50+110+80x+32=256.$$

$$CPI'_{L/S}=x=\frac{256-192}{80}=0.8, \frac{CPI_{old}}{CPI_{new}}=\frac{4}{0.8}=5.$$

(c)

$$CPI'_{FP}=0.6, CPI'_{INT}=0.6, CPI'_{L/S}=2.8, CPI'_{branch}=1.4.$$

$$C'=[50(0.6)+110(0.6)+80(2.8)+16(1.4)]\times 10^6=342.4\times 10^6.$$

$$T'=\frac{342.4\times 10^6}{2\times 10^9}=0.1712s.$$

$$時間減少率=\frac{0.256-0.1712}{0.256}\times 100\%=33.125\%.$$

$$S=\frac{0.256}{0.1712}\approx 1.4953.$$

6.

(a) R-type：sub $v1, $v1, $v0

op rs rt rd shamt funct
000000 00011 00010 00011 00000 100010
0x00621822

(b) I-type：lw $v0, 4($at)

op rs rt immediate
100011 00001 00010 0000000000000100
0x8C220004

7.

(a) $\$s2 = 0 + 10(2) = 20$.

(b)

```
while ((temp = (i > 0))) {
    i--;
    B += 2;
}
```

(c) $I(N) = 5N + 2$.

8.

```
for (i = 0; i < 100; i++) {
    result += MemArray[i];
}
```

result 保留初值。

9.

(a) byte offset $= (4j)(4) = 16j$. Delayed branching：關閉。

```
        addi $t0, $zero, 0
        beq  $s0, $zero, DONE
OUTER:  addi $t1, $zero, 0
        beq  $s1, $zero, NEXT
INNER:  sll  $t2, $t1, 4
        add  $t2, $s2, $t2
        add  $t3, $t0, $t1
        sw   $t3, 0($t2)
        addi $t1, $t1, 1
        bne  $t1, $s1, INNER
NEXT:   addi $t0, $t0, 1
        bne  $t0, $s0, OUTER
DONE:
```

9. (b)

```
.data
D:      .word 0:16
.text
.globl main
main:
        addi $s0, $zero, 3
        addi $s1, $zero, 4
        la   $s2, D
        addi $t0, $zero, 0
        beq  $zero, $zero, TEST1
LOOP1:  addi $t1, $zero, 0
        beq  $zero, $zero, TEST2
LOOP2:  add  $t3, $t0, $t1
        sll  $t2, $t1, 4
        add  $t2, $t2, $s2
        sw   $t3, 0($t2)
        addi $t1, $t1, 1
TEST2:  slt  $t2, $t1, $s1
        bne  $t2, $zero, LOOP2
        addi $t0, $t0, 1
TEST1:  slt  $t2, $t0, $s0
        bne  $t2, $zero, LOOP1
        addi $v0, $zero, 10
        syscall
```